



Impact of Environmental, Social, and Governance Practices on Performance and Value: Evidence from Indonesian Consumer Sectors

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Abstract: The impact of environmental, social, and governance practices on firm performance and firm value remains contested, particularly in emerging markets where sustainability adoption is still evolving. In this study, the relationship between environmental, social, and governance practices, firm performance, and firm value was empirically examined, with a comparative focus on consumer non-cyclical and consumer cyclical sectors in Indonesia. A total of 298 firms listed on the Indonesia Stock Exchange in 2023 were analyzed, comprising 132 consumer non-cyclical and 166 consumer cyclical firms. Environmental, social, and governance performance was operationalized using disclosure-based indicators derived from the Global Reporting Initiative standards. A structural equation modeling approach based on the generalized structured component analysis was employed. Measurement model evaluation was conducted and overall model fit and structural relationships were assessed. Following the removal of invalid indicators, all constructs satisfied validity and reliability requirements, and acceptable model fit was achieved across both sectors. However, the results indicate that environmental, social, and governance practices do not have a statistically significant effect on either firm performance or firm value in both sectors. These findings suggest that environmental, social, and governance implementation in Indonesian consumer sectors remains at an early stage, where disclosure practices have not yet translated into measurable economic outcomes. Furthermore, caution should be exercised by investors when interpreting environmental, social, and governance disclosures as indicators of short-term financial performance. This study contributes to the environmental, social, and governance literature by providing evidence from an emerging market context and highlights the need for more substantive and performance-oriented sustainability integration.

Keywords: Environmental, social, and governance; Firm performance; Firm value; Comparative analysis; Consumer non-cyclicals; Consumer cyclicals; Indonesia

1. Introduction

Environmental, social, and governance performance has rapidly gained prominence as a non-financial dimension of corporate evaluation that potentially shapes both firm performance and firm value. Environmental, social, and governance practices reflect a firm's commitment to sustainability, encompassing environmental stewardship, social responsibility, and effective governance practices. As stakeholder interests broaden beyond traditional financial outcomes, investors, regulators, and analysts increasingly consider environmental, social, and governance disclosure and performance as critical indicators of long-term viability and risk management capacity (Tamasiga et al., 2024). The relationship between environmental, social, and governance practices, performance, and firm value sits at the intersection of stakeholder theory and signaling theory. Stakeholder theory posits that firms creating value for a broader set of stakeholders, not just shareholders, can achieve sustainable competitive advantages (Chipimo et al., 2025). Signaling theory further suggests that transparent environmental, social, and governance disclosure reduces information asymmetry between firms and capital market participants, influencing valuation metrics such as Tobin's Q and stock returns. Empirical studies present mixed evidence on this relationship. In Indonesian contexts, research shows that environmental, social, and governance disclosure

positively influences firm value, especially when firms address information asymmetry through transparent reporting (Angir & Weli, 2024).

Conversely, some evidence suggests that environmental, social, and governance initiatives have a negative effect on firm value unless accompanied by robust firm performance (Sari & Valdiansyah, 2025). Within the Association of Southeast Asian Nations, environmental, social, and governance performance is also seen to positively influence both firm performance and market valuation, highlighting environmental performance as a significant factor within the regional product market (Fadila & Rakhmatullah, 2025). Beyond individual empirical studies, systematic reviews and meta-analyses indicate nuanced dynamics between environmental, social, and governance practices and financial outcomes. A comprehensive open access review shows that strong environmental, social, and governance disclosures correlate with improved firm performance in sectors such as food and retail, while in developing economies the linkage between environmental, social, and governance practices and financial metrics may be weak or insignificant due to contextual factors such as regulatory enforcement and market perception.

Research outside Indonesia affirms that comprehensive environmental, social, and governance performance generally enhances firm valuation, as firms with higher environmental, social, and governance metrics face lower idiosyncratic risk, enjoy improved stakeholder reputation, and may attract long-term capital, particularly in emerging markets (Bashir et al., 2023). Empirical designs from China also demonstrate that environmental, social, and governance performance correlates positively with firm value and that this effect can be mediated by factors such as research and development investment, indicating that strategic investments amplify environmental, social, and governance practices' positive impact (Zhou, 2025). Many studies have investigated environmental, social, and governance practices and firm value independently, and fewer have examined the sectoral differences between industries with varying demand sensitivity, such as consumer non-cyclicals and consumer cyclicals. Sectoral differences may influence how investors respond to environmental, social, and governance investments. Defensive sectors, particularly the consumer non-cyclical sector, are more likely to benefit from stable and long-term environmental, social, and governance signals. In contrast, the consumer cyclical sector may face concerns about the short-term costs associated with environmental, social, and governance initiatives. Although some evidence from the emerging markets of the Association of Southeast Asian Nations exists, direct comparisons between these sectors within a single emerging market, such as Indonesia, remain limited.

To address these gaps, this research analyzes the influence of environmental, social, and governance performance on both firm performance and firm value across consumer non-cyclical and consumer cyclical firms listed on the Indonesian Stock Exchange. By employing a variance-based structural modeling approach, this research aims to provide sector-specific empirical evidence from an emerging market context, contributing to the theoretical understanding and offering practical implications for corporate governance, investor decision-making, and sustainability policy.

2. Methodology

This study adopted a census method covering all listed firms in the consumer non-cyclical and consumer cyclical sectors on the Indonesia Stock Exchange, with a sample of 132 and 166 firms respectively. Environmental, social, and governance data were derived from sustainability reports using 97 disclosure indicators based on Global Reporting Initiative standards. Disclosed indicators were scored 1 and undisclosed ones 0, and total scores represented each firm's disclosure level. Data on firm performance and firm value were collected from annual reports using selected financial indicators. Before analysis, all relevant values were grouped into five classes and converted to a five-point Likert scale to standardize measurements and ensure comparability among variables.

After the variables were prepared and standardized, the data were analyzed using the generalized structured component analysis. The analysis is an approach in structural equation modeling that is variance-based and not bound by the assumption of multivariate normal distribution (Fitriani et al., 2020). The generalized structured component analysis was chosen because it has several advantages, including the ability to handle complex data, flexibility in modeling relationships between variables, and does not require strict normal distribution assumptions. Thus, the generalized structured component analysis provides an opportunity to understand how environmental, social, and governance practices affect the performance and value of consumer non-cyclical and consumer cyclical firms in Indonesia effectively. Several steps below were taken in conducting analysis using the generalized structured component analysis method (Suhriani & Abdurakhman, 2019):

Step 1: Models were developed based on concepts and theories to design structural models and measurement models.

Step 2: Path diagrams that illustrate the relationship patterns between latent variables and their indicators were compiled.

Step 3: Path diagrams were converted into equations.

Step 4: Parameters were calculated that include weight estimates, factor loading estimates, path coefficient estimates, and bootstrap standard error estimates.

Step 5: Parameter coefficients (standard error) and critical ratio statistical values were determined by applying the bootstrap method.

Step 6: The significance of parameters in the structural model was tested.

Step 7: The overall goodness-of-fit of the model to be used was determined.

Step 8: Conclusions were drawn from the results obtained.

Figure 1 shows the structural equation modeling procedure using the generalized structured component analysis (Suhriani & Abdurakhman, 2019).

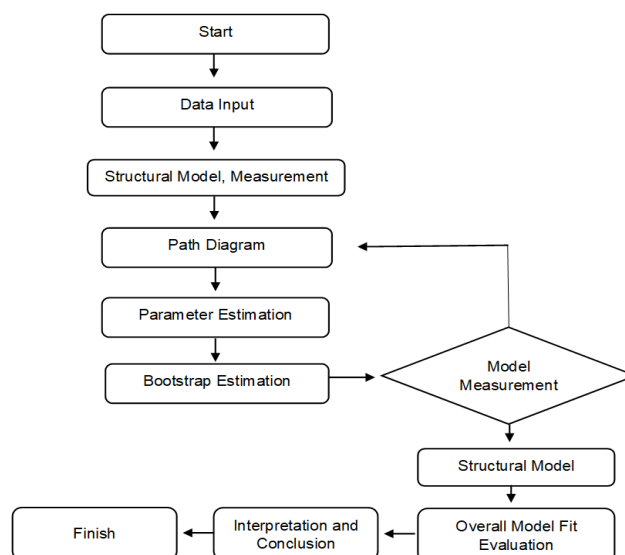


Figure 1. Structural equation modeling procedure using the generalized structured component analysis

This research used the generalised structured component analysis (GSCA) Pro Windows 1.2.1.0 software, with the following component details presented in Table 1.

The analysis data on the generalized structured component analysis components were sourced from secondary data on consumer non-cyclical and consumer cyclical firms, which is available on the Indonesia Stock Exchange website and the websites of each firm. In more detail, the following steps of the generalized structured component analysis were taken in this research:

Step 1: The data required for the generalized structured component analysis components, both from consumer non-cyclical and consumer cyclical firms, were carefully compiled.

Step 2: The collected data were summed up, inputted into intervals, and converted to a 5-point Likert scale in Microsoft Excel and grouped based on analysis requirements.

Step 3: After the data were integrated into the Excel format, the generalized structured component analysis model in the GSCA Pro Windows 1.2.1.0 software was created.

Step 4: The model was created within the software using the tabulated data and was entered through the “New Project” menu.

Step 5: The path diagram of the environmental, social, and governance practices, firm performance, and firm value components was arranged in both sectors.

Step 6: The indicator estimates were arranged for each component.

Step 7: The environmental, social, and governance components were connected to firm performance and firm value using the “Add Path” tool, thereby forming the initial framework of the generalized structured component analysis model for consumer non-cyclical and consumer cyclical firms (Figure 2).

Step 8: After the model is formed, the data tabulated in Excel were run in the “run” submenu to test the measurement model of each component.

Step 9: When the measurement model results for each firm showed invalid components, the path diagram preparation and subsequent steps were repeated until the test results showed that all components in the analysis were valid. In this process, a model that best reflected the characteristics of each firm was established.

Step 10: An overall model fit evaluation was conducted.

Step 11: The analysis results were interpreted and conclusions were drawn from the generalized structured component analysis for each firm.

Table 1. Components of the generalized structured component analysis

No.	Indicator	Parameter	Scale
1	Environmental	- Materials	Ratio scale
		- Energy	
1	Environmental	- Water and effluents	Ratio scale
		- Biodiversity	
1	Environmental	- Emissions	Ratio scale
		- Waste	
1	Environmental	Environmental assessment of suppliers	Ratio scale
		- Employment	
2	Social	- Labor/Management relations	Ratio scale
		- Occupational health and safety	
2	Social	- Training and education	Ratio scale
		- Diversity and equal opportunity	
2	Social	- Non-discrimination	Ratio scale
		Freedom of association and collective bargaining	
2	Social	- Child labor	Ratio scale
		- Forced or compulsory labor	
2	Social	- Security practices	Ratio scale
		- Indigenous peoples' rights	
2	Social	- Local communities	Ratio scale
		- Supplier social assessment	
2	Social	- Public policy	Ratio scale
		- Customer health and safety	
2	Social	- Marketing and labeling	Ratio scale
		- Customer privacy	
3	Governance	Organizational reporting practices	Ratio scale
		- Activities and workforce	
3	Governance	- Governance	Ratio scale
		- Strategy, policies, and practices	
3	Governance	- Stakeholder engagement	Ratio scale
		- Net income	
4	Firm performance	- Total assets	Ratio scale
		- Total equity	
4	Firm performance	- Number of employees separated	Ratio scale
		- Total number of employees	
5	Firm value	- Total market capitalization	Ratio scale
		- Total debt	
5	Firm value	- Market price per share	Ratio scale
		- Book value per share	
5	Firm value	- Earnings per share	Ratio scale
		- Earnings per share	

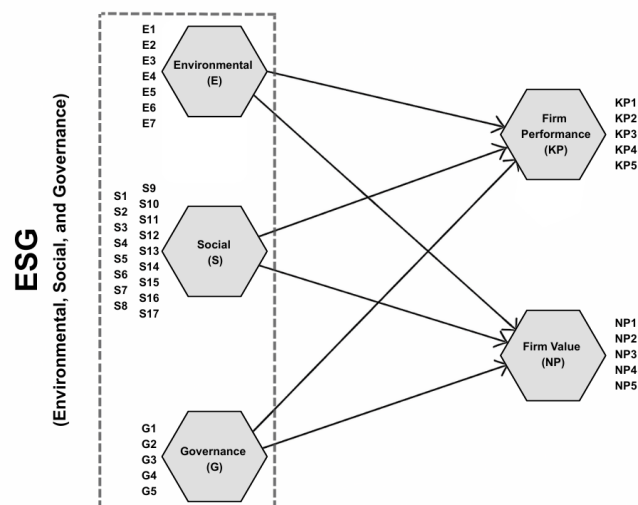


Figure 2. Analysis framework of the generalized structured component analysis model

In Figure 2, E1–E7 represent the environmental indicators of materials, energy, water and effluents, biodiversity, emissions, waste, and supplier environmental assessment, respectively. S1–S17 represent the social indicators of employment, labor/management relations, occupational health and safety, training and education, diversity and equal opportunity, non-discrimination, freedom of association and collective bargaining, child labor, forced or compulsory labor, security practices, indigenous peoples' rights, local communities, supplier social assessment, public policy, customer health and safety, marketing and labeling, and customer privacy, respectively. G1–G5 represent the governance indicators of organizational reporting practices, activities and workforce, governance structure, strategy, policies, and practices, and stakeholder engagement, respectively. KP1–KP5 represent the firm performance indicators of net income, total assets, total equity, number of employees separated, and total number of employees, respectively. NP1–NP5 represent the firm value indicators of total market capitalization, total debt, market price per share, book value per share, and earnings per share, respectively.

3. Results

According to Hwang et al. (2023), the generalized structured component analysis program involves three main stages of analysis: the assessment of the measurement model, the evaluation of the model's overall goodness-of-fit, and the assessment of the structural model. This section presents a detailed explanation of the results obtained at each stage of the generalized structured component analysis for both sectors, described in a clear and systematic manner below.

3.1 Measurement Model

In evaluating the measurement model, convergent validity was assessed based on the loading factor values of each indicator obtained from the generalized structured component analysis. According to the study by Susetyo & Fitrianto (2022), the loading factor value is a representation of the assessment of convergent validity, which can be categorized as good if the value is ≥ 0.70 . Table 2 shows the results of the initial and final validity tests for each sector.

In evaluating the measurement model for the consumer non-cyclical sector, the validity test results show that some indicators are invalid because they have loading factors of less than 0.70. Those invalid indicators include material (E1), supplier environmental assessment (E7), labor/management relations (S2), safety practices (S10), indigenous peoples' rights (S11), supplier social assessment (S13), public policy (S14), customer privacy (S17), number of employees separated (KP4), total number of employees (KP5), total market value of shares (NP1), and book value per share (NP4). Therefore, these indicators must be selected or removed gradually so that all indicators in the validity test are valid.

After the selection process of indicators that were declared invalid in the early stages, the final results of the indicator validity test in the consumer non-cyclical sector can be seen in Table 3.

After invalid indicators were gradually removed, all other indicators in the consumer non-cyclical sector exceeding a value of 0.70 were declared valid. In addition to the consumer non-cyclical sector, Table 4 shows the results of the initial validity test in the consumer cyclical sector.

In early stages of the model evaluation process in the consumer cyclical sector, the validity test results show that some indicators are invalid, including materials (E1), biodiversity (E4), supplier environmental assessment (E7), labor/management relations (S2), occupational health and safety (S3), freedom of association and collective bargaining (S7), security practices (S10), indigenous peoples' rights (S11), supplier social assessment (S13), public policy (S14), marketing and labeling (S16), customer privacy (S17), net income (KP1), number of employees separated (KP4), total debt (NP2), book value per share (NP4), and earnings per share (NP5). Similar to the consumer non-cyclical sector, the invalidity is determined based on the loading factor values of these indicators, which are less than 0.70. Therefore, these indicators must be selected or removed gradually so that all indicators in the validity test are valid.

After the selection process, indicators that were declared invalid in the early stages were removed. The final results of the indicator validity test in the consumer cyclical sector can be seen in Table 5.

After invalid indicators were gradually removed, all indicators in the consumer cyclical sector exceeding a value of 0.70 were declared valid. Based on Table 3 and Table 5, the indicators of each variable in the consumer non-cyclical and consumer cyclical sectors that are declared valid or invalid show differences in terms of both quantity and composition. According to the study by Khalid et al. (2024) in the generalized structured component analysis, the decision on whether an indicator is valid is determined by how strongly it contributes to representing the variable being measured, which is statistically indicated by the loading factor or indicator weight value. Indicators with higher loading values show a better ability to reflect the research variable, while indicators with low loading values show a weak contribution and are therefore not declared valid in the measurement model. In this case, indicators with weak contributions can be triggered by limitations in data variation, sector relevance, and limited disclosure properties, which are declared invalid.

Table 2. Initial validity test results for the consumer non-cyclical sector

No.	Variable	Indicator	Loading	Result
1	Environmental (E)	E1	0.620	Not valid
		E2	0.929	Valid
		E3	0.914	Valid
		E4	0.78	Valid
		E5	0.918	Valid
		E6	0.938	Valid
		E7	0.600	Not valid
2	Social (S)	S1	0.833	Valid
		S2	0.265	Not valid
		S3	0.872	Valid
		S4	0.873	Valid
		S5	0.902	Valid
		S6	0.809	Valid
		S7	0.731	Valid
		S8	0.829	Valid
		S9	0.833	Valid
		S10	0.273	Not valid
		S11	0.522	Not valid
		S12	0.862	Valid
		S13	0.638	Not valid
		S14	0.165	Not valid
		S15	0.829	Valid
		S16	0.724	Valid
		S17	0.384	Not valid
3	Governance (G)	G1	0.982	Valid
		G2	0.991	Valid
		G3	0.976	Valid
		G4	0.988	Valid
		G5	0.993	Valid
4	Firm performance (KP)	KP1	0.926	Valid
		KP2	0.875	Valid
		KP3	0.937	Valid
		KP4	0.180	Not valid
		KP5	0.608	Not valid
5	Firm value (NP)	NP1	0.617	Not valid
		NP2	0.747	Valid
		NP3	0.874	Valid
		NP4	-0.05	Not valid
		NP5	0.834	Valid

Table 3. Final validity test results for the consumer non-cyclical sector

No.	Variable	Indicator	Loading	Result
1	Environmental (E)	E2	0.945	Valid
		E3	0.924	Valid
		E4	0.767	Valid
		E5	0.935	Valid
		E6	0.952	Valid
2	Social (S)	S1	0.845	Valid
		S3	0.871	Valid
		S4	0.88	Valid
		S5	0.908	Valid
		S6	0.803	Valid
		S7	0.712	Valid
		S8	0.84	Valid
		S9	0.848	Valid
		S12	0.863	Valid
		S15	0.837	Valid
3	Governance (G)	S16	0.756	Valid
		G1	0.982	Valid
		G2	0.991	Valid
		G3	0.976	Valid
		G4	0.988	Valid

		G5	0.993	Valid
4	Firm performance (KP)	KP1	0.928	Valid
		KP2	0.928	Valid
		KP3	0.949	Valid
5	Firm value (NP)	NP3	0.981	Valid
		NP5	0.976	Valid

Table 4. Initial validity test results for the consumer cyclical sector

No.	Variable	Indicator	Loading	Result
1	Environmental (E)	E1	0.372	Not valid
		E2	0.938	Valid
		E3	0.761	Valid
		E4	0.535	Not valid
		E5	0.853	Valid
		E6	0.921	Valid
		E7	0.501	Not valid
2	Social (S)	S1	0.84	Valid
		S2	0.373	Not valid
		S3	0.667	Not valid
		S4	0.928	Valid
		S5	0.834	Valid
		S6	0.752	Valid
		S7	0.291	Not valid
		S8	0.835	Valid
		S9	0.835	Valid
		S10	0.0	Not valid
		S11	0.0	Not valid
		S12	0.834	Valid
		S13	0.472	Not valid
		S14	0.018	Not valid
		S15	0.764	Valid
		S16	0.633	Not valid
		S17	0.67	Not valid
3	Governance (G)	G1	0.989	Valid
		G2	0.986	Valid
		G3	0.979	Valid
		G4	0.992	Valid
		G5	0.984	Valid
4	Firm performance (KP)	KP1	0.352	Not valid
		KP2	0.855	Valid
		KP3	0.847	Valid
		KP4	0.08	Not valid
		KP5	0.758	Valid
5	Firm value (NP)	NP1	0.814	Valid
		NP2	0.188	Not valid
		NP3	0.874	Valid
		NP4	0.097	Not valid
		NP5	-0.115	Not valid

In addition, differences in the number and composition of valid indicators between the consumer non-cyclical and consumer cyclical sectors also reflect differences in data characteristics and disclosure patterns. Therefore, the strength of the relationship between measurement indicators and the research variables they represent is not always the same in each sector. These findings are in line with the results of research by Amanah & Fitri (2023), which show that in the generalized structured component analysis, not all measurement indicators always have the same strong contribution in representing the research variable. Indicators with low weight or loading values reflect weak empirical relevance to the measured variable, so these indicators are not retained in the measurement model.

In the consumer non-cyclical sector, valid environment indicators include E2, E3, E4, E5, and E6, while in the consumer cyclical sector, valid indicators include E2, E3, E5, and E6, with E4 being unconfirmed. This shows that environmental indicators in the consumer non-cyclical sector are represented by more diverse indicators than those in the consumer cyclical sector. In terms of social indicators, the consumer non-cyclical sector has a greater number of valid indicators, namely S1, S3, S4, S5, S6, S7, S8, S9, S12, S15, and S16. Meanwhile, the consumer cyclical sector only includes S1, S4, S5, S6, S8, S9, S12, S15, and S17. This difference shows that the representation of social aspects in the consumer non-cyclical and consumer cyclical sectors is formed by indicators that are not

entirely the same. Meanwhile, in terms of governance indicators, both sectors show relatively similar results, where all indicators, G1 to G5, are declared valid with very high loading values. In the firm performance (KP) variable, the consumer non-cyclical sector is represented by the KP1, KP2, and KP3 indicators, while the consumer cyclical sector is represented by KP2, KP3, and KP5. In the firm value (NP) variable, NP3 and NP5 are identified as valid indicators for the consumer non-cyclical sector, whereas NP1 and NP3 are valid indicators for the consumer cyclical sector.

Table 5. Final validity test results for the consumer cyclical sector

No.	Variable	Indicator	Loading	Result
1	Environmental (E)	E2	0.925	Valid
		E3	0.81	Valid
		E5	0.867	Valid
		E6	0.924	Valid
2	Social (S)	S1	0.847	Valid
		S4	0.902	Valid
		S5	0.876	Valid
		S6	0.789	Valid
		S8	0.849	Valid
		S9	0.849	Valid
		S12	0.81	Valid
		S15	0.769	Valid
3	Governance (G)	S17	0.731	Valid
		G1	0.989	Valid
		G2	0.986	Valid
		G3	0.979	Valid
		G4	0.992	Valid
4	Firm performance (KP)	G5	0.984	Valid
		KP2	0.862	Valid
		KP3	0.856	Valid
5	Firm value (NP)	KP5	0.772	Valid
		NP1	0.837	Valid
		NP3	0.874	Valid

Thus, the differences in indicators declared valid between the consumer non-cyclical and consumer cyclical sectors show that both sectors are represented by different combinations of indicators, even though they remain within the same construct. This confirms that the measurement of variables in consumer non-cyclical and consumer cyclical sectors is adjusted to the characteristics of each sector's data. Therefore, the validity test results reflect the specific measurement structure in each sector. During the evaluation of the measurement model, several indicators were removed because their loading values did not meet the recommended threshold. In total, 13 indicators were excluded in the consumer non-cyclical sector and 16 indicators were removed in the consumer cyclical sector through an iterative model re-estimation process. The elimination of indicators with low loading values is a common procedure in component-based structural equation modeling to improve indicator reliability and ensure that each retained indicator contributes meaningfully to the latent construct.

Importantly, the removal of these indicators did not alter the conceptual structure of the constructs used in this research. In component-based structural equation modeling approaches such as the generalized structured component analysis, as discussed by Sarstedt & Hwang (2020), latent variables are conceptualized as weighted composites of their indicators rather than as common factors. Consequently, indicators with weak empirical contributions may be removed without affecting the theoretical interpretation of the construct, provided that the remaining indicators still adequately represent the conceptual domain of the variable. Furthermore, all constructs in the final measurement model retain at least two valid indicators. In structural equation modeling, constructs represented by two or more indicators remain statistically identifiable and can be reliably estimated when the indicators demonstrate sufficient loading values and conceptual relevance. Therefore, despite the elimination of several low-loading indicators, the final measurement model continues to capture the theoretical meaning of the environmental, social, and governance practices, firm performance, and firm value constructs analyzed in this research. Figure 3 and Figure 4 show the valid generalized structured component analysis model framework for the two sectors.

After all valid indicators were identified, a reliability test was conducted to measure the proportion of variance explained and Dillon-Goldstein's rho (composite reliability) values. Based on the research conducted by Rudianto et al. (2025), to demonstrate convergent validity and composite reliability, the proportion of variance explained values must be ≥ 0.50 and rho values must be ≥ 0.70 to be considered reliable. The results of the reliability test for each sector are shown below.

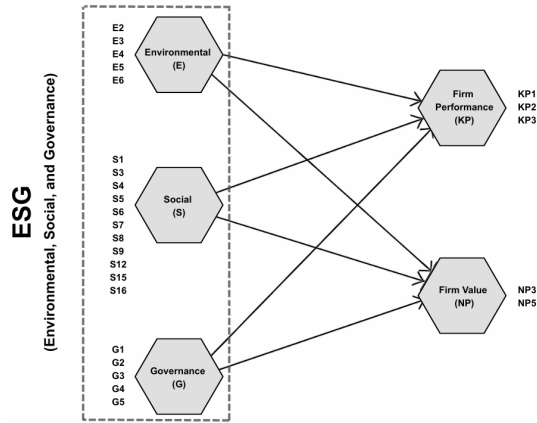


Figure 3. Valid generalized structured component analysis model framework for the consumer non-cyclical sector

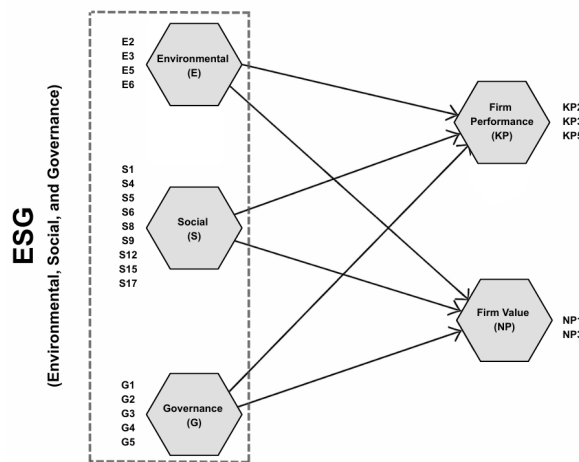


Figure 4. Valid generalized structured component analysis model framework for the consumer cyclical sector

Based on the analysis results shown in Table 6 and Table 7, the proportion of variance explained and rho values of each variable in both sectors meet the reliability criteria, namely proportion of variance explained ≥ 0.50 and $\rho \geq 0.70$. Therefore, the analysis evaluation stage can proceed to the next stage, which is the evaluation of the model goodness-of-fit criteria.

Table 6. Reliability test results for the consumer non-cyclical sector

No.	Criteria	Environmental (E)	Social (S)	Governance (G)	Firm Value (NP)	Firm Performance (KP)
1	Proportion of variance explained	0.824	0.697	0.972	0.957	0.875
2	rho	0.959	0.962	0.994	0.978	0.954

Table 7. Reliability test results for the consumer cyclical sector

No.	Criteria	Environmental (E)	Social (S)	Governance (G)	Firm Value (NP)	Firm Performance (KP)
1	Proportion of variance explained	0.78	0.683	0.972	0.732	0.69
2	rho	0.934	0.951	0.994	0.845	0.87

3.2 Evaluation Criteria for the Goodness-of-Fit Model

The determination of goodness-of-fit in the generalized structured component analysis includes observation of a number of evaluation criteria, such as the overall fit index, adjusted fit index, structural model fit index, measurement model fit index, goodness-of-fit index, and standardized root mean square residual. Cho & Hwang

(2024) explained that the overall fit index value ranges from 0 to 1, where there are no specific restrictions for the overall fit index in showing model fit adequacy. Regarding the goodness-of-fit index and standardized root mean square residual values, Cho et al. (2020) suggested that an acceptable fit can be accepted if the goodness-of-fit index value is ≥ 0.93 or the standardized root mean square residual value is ≤ 0.08 . The model fit measure results for each sector are presented below.

Based on the results shown in Table 8 and Table 9, it can be concluded that the models of both sectors show a good level of conformity and meet the acceptable fit criteria. Therefore, the analysis evaluation stage can proceed to the structural model evaluation stage (inner model).

Table 8. Model fit measures for the consumer non-cyclical sector

Overall Fit Index	Adjusted Fit Index	Structural Model Fit Index	Measurement Model Fit Index	Goodness-of-Fit Index	Standardized Root Mean Square Residual	Overall Prediction Error	Overall Prediction Error (Structural Model)	Overall Prediction Error (Measurement Model)
0.687	0.682	0.022	0.815	0.992	0.056	0.33	1.012	0.199

Table 9. Model fit measures for the consumer cyclical sector

Overall Fit Index	Adjusted Fit Index	Structural Model Fit Index	Measurement Model Fit Index	Goodness-of-Fit Index	Standardized Root Mean Square Residual	Overall Prediction Error	Overall Prediction Error (Structural Model)	Overall Prediction Error (Measurement Model)
0.634	0.629	0.021	0.768	0.986	0.067	0.386	1.033	0.245

3.3 Structural Model Evaluation (Inner Model)

The evaluation of the structural model (inner model) was conducted by reviewing the path coefficient values and their significance levels, which can be observed through the output of data processing using the generalized structured component analysis. The structural model evaluation aims to assess the influence between variables, where the significance of the influence between variables is tested based on the critical ratio value resulting from the division between the parameter estimate and the standard error. If the critical ratio value is greater than 1.96 or less than -1.96, then the relationship can be declared significant at a 5% two-tailed significance level in accordance with the commonly used statistical testing standards (Hair et al., 2021). The path coefficients for both sectors based on the generalized structured component analysis are shown below.

Based on the path coefficients shown in Table 10 and Table 11, it can be seen that environmental, social, and governance practices in both sectors have no significant effect on firm performance or firm value, as they are below the statistical significance threshold. None of the environmental, social, and governance dimensions demonstrate statistically significant relationships with firm performance or firm value in either the consumer non-cyclical or consumer cyclical sectors. Although several path coefficients show positive or negative directions, the corresponding critical ratio values remain below the statistical significance threshold. This indicates that the environmental, social, and governance variables included in the research model are not able to explain variations in firm performance or firm value in the two sectors during the observation period. To provide a clearer overview of these structural relationships, Table 12 summarizes the statistical significance of each relationship between environmental, social, and governance dimensions and firm outcomes in both sectors.

The relationship between environmental factors and firm value (E→NP) has a positive coefficient of 0.268, but a critical ratio value of 1.24 indicates that this effect is not yet statistically significant. A similar pattern is observed in the relationship between environmental factors and firm performance (E→KP), with a positive coefficient of 0.375 and a critical ratio of 1.37. Although the direction of the relationship is positive, these findings indicate that environmental practices and disclosures in the consumer non-cyclical sector are not yet strong or consistent enough to have a measurable impact on firm performance or value.

Meanwhile, social and governance indicators in the consumer non-cyclical sector show relatively small coefficients, some of which are negative in relation to firm performance and firm value. The relationship between social factors and firm value (S→NP) and social factors and firm performance (S→KP) have critical ratio values of -0.04 and 0.81, respectively, while the relationship between governance factors and firm value (G→NP) and firm performance (G→KP) have critical ratio values of -1.34 and -1.74. Although the relationship between governance and firm performance is close to the significance threshold, the value still does not meet the statistical criteria. Therefore, it can be concluded that environmental, social, and governance practices have no significant effect on the performance and value of firms in the consumer non-cyclical sector.

Table 10. Path coefficients for the consumer non-cyclical sector

Variable	Estimate	Standard Error	Critical Ratio
Environmental (E)→Firm value (NP)	0.268	0.217	1.24
Social (S)→Firm value (NP)	-0.009	0.203	-0.04
Governance (G)→Firm value (NP)	-0.198	0.148	-1.34
Environmental (E)→Firm performance (KP)	0.375	0.274	1.37
Social (S)→Firm performance (KP)	0.247	0.304	0.81
Governance (G)→Firm performance (KP)	-0.352	0.202	-1.74

Table 11. Path coefficients for the consumer cyclical sector

Variable	Estimate	Standard Error	Critical Ratio
Environmental (E)→Firm value (NP)	0.31	0.28	1.11
Social (S)→Firm value (NP)	0.05	0.388	0.13
Governance (G)→Firm value (NP)	-0.107	0.334	-0.32
Environmental (E)→Firm performance (KP)	-0.072	0.121	-0.60
Social (S)→Firm performance (KP)	0.293	0.246	1.19
Governance (G)→Firm performance (KP)	-0.075	0.169	-0.44

Table 12. Structural model relationships

Relationship	Consumer Non-Cyclical Sector (Critical Ratio)	Consumer Cyclical Sector (Critical Ratio)	Result
Environmental (E)→Firm value (NP)	1.24	1.11	Not significant
Environmental (E)→Firm performance (KP)	1.37	-0.60	Not significant
Social (S)→Firm value (NP)	-0.04	0.13	Not significant
Social (S)→Firm performance (KP)	0.81	1.19	Not significant
Governance (G)→Firm value (NP)	-1.34	-0.32	Not significant
Governance (G)→Firm performance (KP)	-1.74	-0.44	Not significant

Note: Relationships are considered statistically significant when the critical ratio value exceeds ± 1.96

In the consumer non-cyclical sector, the insignificant impact of social and governance indicators on firm performance and firm value can be explained by the characteristics of food and agribusiness-based firms, which are more focused on the sustainability of raw material supplies, production stability, and smooth distribution. Therefore, firm performance is largely determined by the ability to maintain relationships with agricultural suppliers and supply chain continuity (Wedari & Alfian, 2024). In addition, social and governance policies in this sector are generally at a relatively uniform level of compliance so that the differences between firms are not yet prominent enough to be reflected in firm performance or firm value.

The consumer cyclical sector has relatively similar results, as shown in Table 11. All environmental, social, and governance influence paths on firm performance and firm value show that critical ratio values are well below the significance threshold of 1.96. The relationship between environmental factors and firm value (E→NP) has a positive coefficient of 0.31 with a critical ratio of 1.11, while environmental factors and firm performance (E→KP) show a negative coefficient of -0.072 with a critical ratio of -0.60. These findings indicate that in the consumer cyclical sector, the implementation of environmental aspects does not have a consistent pattern of influence on operational performance or market perception.

Social indicators in the consumer cyclical sector show positive coefficients for both firm value (0.05) and firm performance (0.293), but each has a critical ratio value of 0.13 and 1.19, respectively, indicating that the effect is not yet statistically significant. Similarly, governance indicators show negative coefficients for firm performance and firm value with relatively low critical ratio values (-0.32 and -0.44). Overall, these results confirm that the implementation and disclosure of environmental, social, and governance practices in the consumer cyclical sector has not been able to explain variations in firm performance and firm value significantly in this research model. The insignificance of the social and governance indicators shows that firm performance and firm value in this sector are more influenced by fluctuations in demand, consumer preferences, and macroeconomic conditions. This aligns with the results found by Wang et al. (2026), which suggest that social and governance practices also tend to follow the same regulatory standards, so they have not formed a strong enough differentiation to influence firm performance and firm value.

Therefore, it can be concluded that the insignificant impact of environmental, social, and governance practices on firm performance and firm value is generally caused by the limited role of those practices in consumer non-

cyclical and consumer cyclical firms in Indonesia. This condition can be attributed to the relatively low level of environmental, social, and governance disclosure, which has not yet reached the maximum disclosure standards with disclosure levels of only 38% in the consumer non-cyclical sector and 14% in the consumer cyclical sector. Therefore, the available environmental, social, and governance information is not yet strong enough to influence managerial decision-making or investor perceptions. In addition, these findings also indicate the existence of other external factors beyond environmental, social, and governance practices, such as macroeconomic conditions, market structure, industry characteristics, and firm fundamentals, which have a more dominant influence on firm performance and firm value in both sectors.

Comparatively, although both the consumer non-cyclical and consumer cyclical sectors show insignificant environmental, social, and governance influences, the differences in the direction and magnitude of the path coefficients reflect different sectoral characteristics. The consumer non-cyclical sector tends to show more stable and relatively consistent coefficients, particularly for the environmental indicators, while the consumer cyclical sector shows more fluctuating coefficient variations. These findings indicate that environmental, social, and governance sensitivity to firm performance and firm value is greatly influenced by industry characteristics, meaning that the effectiveness of environmental, social, and governance implementation cannot be generalized uniformly across sectors.

In the context of the empirical findings of this research, firms in both sectors still need to improve the quality and strategy of environmental, social, and governance implementation in a more systematic and integrated manner into their core business processes so that its implementation can truly contribute to long-term competitiveness and sustainability. The insignificant effect of environmental, social, and governance practices on firm performance and value indicates that those practices in the consumer non-cyclical and consumer cyclical sectors are still in the early stages of development and do not yet fully function as a mechanism for creating economic value. Broadly speaking, the results of this research show that the consumer non-cyclical sector is a relatively more strategic investment choice in supporting national food security, given its more stable, efficient, and community-oriented characteristics. Conversely, the consumer cyclical sector, which is more influenced by economic fluctuations, tends to face higher demand volatility, thereby limiting the consistency of performance and the effectiveness of environmental, social, and governance practices in supporting sustainable firm value. Thus, sectoral differences are an important aspect in assessing the role of those practices and the contribution of both sectors in supporting sustainable economic development and food systems in Indonesia.

Based on these findings, the results of this research can also be explained through the perspectives of stakeholder theory and signaling theory. Within the framework of stakeholder theory, environmental, social, and governance practices are viewed as instruments for firms to fulfill the interests of various stakeholder groups, including business, social, financial, and supervisory stakeholders, as well as a means of building social legitimacy (Subroto & Endaryati, 2024). However, the insignificant effect of environmental, social, and governance practices on firm performance and value in the consumer non-cyclical and consumer cyclical sectors indicates that these practices carried out by firms have not been fully integrated into their core business strategies. Therefore, the benefits for stakeholders are not yet directly reflected in improved operational performance or firm value.

Furthermore, these results indicate that firms' attention to social and environmental stakeholder interests, as represented by environmental and social indicators, has not been able to generate economic value that can be felt tangibly by the firm (Retnosari et al., 2025). This condition is in line with the findings that in the consumer non-cyclical sector, although environmental indicators show a positive coefficient, the effect is not yet significant, while social and governance indicators show relatively small and unstable coefficients. From a stakeholder theory perspective, this reflects that the firm's relationship with stakeholder groups outside of capital owners has not yet been optimally managed to support the creation of firm value.

Meanwhile, in the consumer cyclical sector, fluctuations in the direction and magnitude of coefficients across all environmental, social, and governance indicators show that firms' attention to stakeholders has not yet resulted in consistent performance patterns and firm value. This indicates that stakeholder management in the consumer cyclical sector, which is heavily influenced by demand dynamics and economic conditions, still faces limitations in driving firm performance and firm value through sustainability practices.

From the perspective of signaling theory, environmental, social, and governance disclosure is essentially expected to serve as a signal regarding the quality of the firm, its sustainability prospects, and the quality of its governance to investors and stakeholders (Huang et al., 2025). However, the low critical ratio values across all environmental, social, and governance influence pathways on firm performance and firm value in both sectors indicate that the environmental, social, and governance information disclosed by firms is not yet functioning effectively as a signal capable of influencing market perceptions. This condition indicates that the environmental, social, and governance signals sent by firms are not yet strong, credible, and consistent enough to differentiate firm quality in the eyes of investors. This is reflected in the findings that, in both the consumer non-cyclical and consumer cyclical sectors, although some coefficients show a positive direction, particularly in environmental indicators, the market has not responded to this information as the main basis for assessing firm performance and firm value.

Furthermore, within the framework of signaling theory, signals can only be effective if they contain sufficient information and are able to reduce information asymmetry between management and investors (Purba, 2023). The findings of this research indicate that the relatively low level of environmental, social, and governance disclosure, which is still far from the maximum disclosure standard, means that the disclosed information is not yet able to serve as a relevant signal for investors. As a result, investment decisions and market assessments are still largely influenced by signals other than environmental, social, and governance practices, such as macroeconomic conditions, industry characteristics, market structure, and firm fundamentals.

Comparatively, although the consumer non-cyclical and consumer cyclical sectors both show insignificant environmental, social, and governance influences, the difference in direction and stability of the path coefficients indicates that the response to environmental, social, and governance signals is greatly influenced by sectoral characteristics. The consumer non-cyclical sector tends to show a more stable relationship pattern, especially in environmental indicators, in line with the sector's character of being oriented towards meeting the basic needs of the community. Conversely, the consumer cyclical sector shows more fluctuating coefficient variations, reflecting that the effectiveness of environmental, social, and governance practices as a signal and as a means of fulfilling stakeholder interests is more influenced by market dynamics and economic conditions.

Thus, based on the integration of stakeholder theory and signaling theory, it can be concluded that environmental, social, and governance practices in consumer non-cyclical and consumer cyclical sector firms in Indonesia are still in the early stages of development and neither fully function as a means of fulfilling stakeholder interests that can create economic value nor as a signal of firm quality for the market. Therefore, improving the quality, consistency, and integration of environmental, social, and governance strategies into core business processes is a prerequisite for environmental, social, and governance practices to play a more effective role in supporting the firm's performance and value in a sustainable manner.

4. Discussion

The results of this research indicate that environmental, social, and governance dimensions do not have a statistically significant effect on firm performance or firm value in either the consumer non-cyclical or consumer cyclical sectors. All path coefficients yield critical ratio values below the 1.96 threshold, indicating that the expected causal relationships are not supported. Although several coefficients, especially those related to the environmental dimension, are positive, they lack sufficient magnitude to demonstrate meaningful economic relevance. The positive but non-significant relationships in this research reflect the current stage of environmental, social, and governance implementation in Indonesian firms, which appears to be compliance-oriented and not fully embedded into strategic and operational frameworks. This interpretation aligns with the result found by Afifah et al. (2025), which shows international evidence that the effects of environmental, social, and governance disclosure on firm outcomes vary significantly across contexts. Their systematic literature review further indicates that empirical findings regarding the relationship between environmental, social, and governance disclosure and firm value remain mixed, with some studies reporting positive associations while others document insignificant or inconsistent effects, particularly when moderating variables such as profitability are not considered.

Supporting this heterogeneous evidence, related research within emerging markets finds that comprehensive environmental, social, and governance information can positively influence market reactions and investor sentiment, but the strength of this effect depends on market conditions, ownership structures, and disclosure quality (Itan et al., 2025). This suggests that environmental, social, and governance information may primarily function as a signal rather than a direct driver of performance, and effectiveness is conditioned by investor interpretation and market dynamics. In contrast to studies reporting positive performance outcomes from environmental, social, and governance practices, there is also empirical evidence consistent with the present findings. For example, research from Tang et al. (2025) demonstrates that increased environmental, social, and governance disclosure may negatively affect financial performance in Indonesia, suggesting that environmental, social, and governance commitments can impose costs that outweigh short-term financial benefits. The results highlight that the relationship between environmental, social, and governance practices and firm outcomes is not universally positive and can be context-specific. Similar mixed findings have also been documented in several recent open-access studies. For example, Makhdalena et al. (2023) showed that environmental, social, and governance disclosure in Southeast Asian firms can improve corporate performance, but the magnitude of the effect varies across industries and institutional contexts. Likewise, Li et al. (2024) reported that environmental, social, and governance disclosure positively influences firm value in Chinese listed firms, although its impact on operational performance indicators remains inconsistent. These findings reinforce the view that the relationship between environmental, social, and governance practices and firm outcomes is highly dependent on contextual and institutional factors.

Additional international evidence supports the notion that industry environment and regional market characteristics influence environmental, social, and governance relevance. Research in emerging markets of the Association of Southeast Asian Nations indicates that environmental, social, and governance performance significantly affects some financial metrics like investment efficiency and return on assets, but may have limited

influence on other indicators like Tobin's Q, depending on moderating factors such as board composition (Nindita & Hanggraeni, 2024). This resonates with the present results, where sectoral differences between consumer non-cyclicals and consumer cyclicals suggest that environmental risk management may be more immediately relevant in essential goods sectors, whereas cyclic sectors dominated by demand volatility show less consistent environmental, social, and governance effects. The findings also correspond with the results found by Angela & Toto (2025), which show that in developing countries, the impact of environmental, social, and governance disclosure does not uniformly enhance financial outcomes unless it is supported by competitive advantages and broader institutional mechanisms. Rudolph & Aliamutu (2026) also highlighted that the unclear role of governance disclosure could thus reflect the limited capacity of environmental, social, and governance reporting to serve as a credible signal of financial strength in the short term, especially when market information environments are underdeveloped. Recent empirical evidence from the markets within the Association of Southeast Asian Nations also highlights similar dynamics. Tripopsakul (2025) found that environmental, social, and governance practices can contribute to financial performance through mechanisms such as green innovation, yet the effect is not always immediate and varies across environmental, social, and governance dimensions. This supports the interpretation that environmental, social, and governance initiatives may require deeper integration into corporate strategy before measurable financial outcomes become evident.

Therefore, this research contributes to the existing environmental, social, and governance literature by providing empirical evidence from an emerging market context using a component-based structural equation modeling approach, which reveals that environmental, social, and governance dimensions do not have a significant direct effect on firm performance and firm value across sectors. By highlighting the non-significant relationships and sectoral differences, this research extends prior research by demonstrating that environmental, social, and governance effects are not universally beneficial and may depend on structural, institutional, and measurement-related conditions. From a theoretical perspective, these results suggest that environmental, social, and governance practices in the consumer non-cyclical and consumer cyclical sectors in Indonesia may still be in the early stages of strategic maturity, where their benefits have not yet been internalized as drivers of financial performance or firm valuation. Although stakeholder theory posits that firms can enhance long-term value through stakeholder engagement, the absence of significant effects indicates that environmental, social, and governance efforts have not yet translated into measurable performance benefits or investor recognition. Similarly, while signaling theory anticipates that environmental, social, and governance disclosures should reduce information asymmetry and improve market expectations, the negligible impact observed implies that current environmental, social, and governance signals may not be sufficiently detailed or credible for investors in these sectors.

These findings are particularly relevant for emerging market contexts, where sustainability reporting frameworks, regulatory enforcement, and investor awareness of environmental, social, and governance issues are still evolving. In such environments, environmental, social, and governance disclosures may initially function more as compliance mechanisms than as strategic instruments for value creation. Overall, this research contributes to the broader environmental, social, and governance literature by reinforcing that the influence of environmental, social, and governance practices is context-dependent and may not be statistically detectable in all settings, particularly where disclosure quality, integration with business strategy, and investor utilization of non-financial information are still evolving. From a practical perspective, these results imply that managers should consider integrating environmental, social, and governance initiatives more deeply into corporate strategy rather than treating them solely as reporting requirements. Investors may also need to evaluate the substantive quality of environmental, social, and governance implementation rather than relying only on disclosure presence, while policymakers can play an important role in strengthening environmental, social, and governance reporting standards and institutional frameworks to enhance the credibility and comparability of sustainability information in emerging markets. Firms in both the consumer non-cyclical and consumer cyclical sectors may therefore benefit from advancing beyond compliance-level reporting toward more strategic environmental, social, and governance integration to foster performance and long-term value creation.

5. Conclusions

This research concludes that environmental, social, and governance dimensions do not have a statistically significant effect on firm performance or firm value in either the consumer non-cyclical or consumer cyclical sectors in Indonesia. Although several environmental coefficients are positive, environmental, social, and governance practices in both sectors have not yet been sufficiently integrated into core business strategies to generate measurable operational or market outcomes. These findings underline the importance of sectoral characteristics in shaping the effectiveness of sustainability implementation and confirm that the economic relevance of environmental, social, and governance practices cannot be generalized across industries in an emerging market context.

The main significance of this research lies in providing sector-specific empirical evidence that environmental, social, and governance disclosure and implementation in Indonesian consumer firms remain at an early and largely

compliance-oriented stage, limiting their ability to function as strategic tools for value creation. A key limitation of this research is that it relies on secondary disclosure data and a cross-sectional design, which restricts the ability to capture the dynamic and long-term effects of environmental, social, and governance initiatives. Future research is therefore encouraged to employ longitudinal designs, incorporate alternative performance and market-based indicators, and explore additional mediating or moderating factors such as innovation capability, supply chain resilience, or corporate risk management to better explain how and when environmental, social, and governance practices can contribute to firm performance and firm value in different industrial contexts.

Author Contributions

Conceptualization, N.N.A.P.R., I.N.G.U., and G.M.K.A.; methodology, N.N.A.P.R., I.N.G.U., and G.M.K.A.; software, N.N.A.P.R.; validation, N.N.A.P.R., I.N.G.U., and G.M.K.A.; formal analysis, N.N.A.P.R.; investigation, N.N.A.P.R.; resources, I.N.G.U. and G.M.K.A.; data curation, N.N.A.P.R.; writing—original draft preparation, N.N.A.P.R.; writing—review and editing, N.N.A.P.R., I.N.G.U., and G.M.K.A.; visualization, N.N.A.P.R.; supervision, I.N.G.U. and G.M.K.A.; project administration, N.N.A.P.R.; funding acquisition, I.N.G.U. and G.M.K.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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